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Hiring Manager
Wilder Systems

Dear Hiring Manager,

I am writing to apply for the Senior Robotics Software Engineer position at Wilder Systems. As a robotics engineer with hands-on experience spanning ROS 2-based motion control, embedded firmware for industrial controllers, and cross-functional team leadership, I bring the full-stack integration depth that deploying autonomous platforms at production scale demands.

On the L.I.F.T. hybrid robot project, I developed a ROS 2 system coupling a physics simulator with a Model Predictive Controller to execute collision-free trajectories with real-time state logging. Separately, I established an OptiTrack-based feedback pose control system that enabled sub-centimeter trajectory tracking and iterative controller refinement. That workflow—simulate, deploy, measure, refine—mirrors the precision motion control and real-time telemetry cycle central to Wilder's drilling and defastening platforms.

Beyond control algorithms, I have direct industrial robotics experience. At Xiaoxian AI Technology, I programmed a six-axis CRP2 robotic arm in embedded C and ladder logic, designed an STM32-based Modbus RTU gateway for real-time PC-to-controller communication, and debugged Siemens S7-1200 PLC programs for a third-party automation workflow. I also led multi-team projects—managing three interdisciplinary teams and a \$20,000 budget during a six-month ROBOCON competition, and coordinating a five-member cross-functional team on the L.I.F.T. platform. These experiences prepared me to bridge mechanical, electrical, and software teams as Wilder scales its WS5000 fleet.

I recognize that MoveIt and AI-based perception are areas where I would need to ramp up quickly; my ROS 2 and sensor integration foundation positions me to do so. I would welcome the opportunity to discuss how my background aligns with Wilder's mission. Thank you for your consideration.

Sincerely,
Felix Hu